3D打印是怎么工作的

从电脑模型到真实物体

How 3D Printing Works

From a computer model to a real object

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Footnotes mark names, festival terms and words above HSK 4 (HSK 3.0 lists: github.com/ivankra/hsk30, MIT).

1 你有没有想过，如果可以像打印一张纸一样“打印”一个杯子、一个零件[1]，甚至一座房子，那会是什么样子？今天，这已经不是梦想了。3D打印技术让人们可以把电脑里的模型直接变成真实的物体[2]。这篇文章简单介绍一下3D打印的基本原理[3]、常见的几种打印方法，以及它的优点和问题。

加法和减法

2 传统的制造方法大多是“减法”。比如说，要做一个金属[4]零件，工人常常先拿一块比零件大的材料，然后用机床[5]切、钻[6]、磨[7]，把不需要的部分去掉[8]，最后剩下[9]的就是想要的形状。这种方法叫做减材制造[10]。

1 零件 *língjiàn* part, component
2 物体 *wùtǐ* object
3 原理 *yuánlǐ* principle
4 金属 *jīnshǔ* metal
5 机床 *jīchuáng* machine tool
6 钻 *zuān* to drill
7 磨 *mó* to grind
8 去掉 *qùdiào* to remove
9 剩下 *shèngxià* to be left over
10 减材制造 *jiǎncái zhìzào* subtractive manufacturing

Have you ever imagined "printing" a cup, a machine part or even a house 1
the way you print a sheet of paper? Today this is no longer a dream. 3D printing technology lets people turn a model on a computer directly into a real object. This article gives a short introduction to the basic principles of 3D printing, the most common printing methods, and its advantages and problems.

Adding and Subtracting

Most traditional manufacturing methods work by "subtraction". For exam- 2
ple, to make a metal part, a worker usually starts with a block of material larger than the part, then cuts, drills and grinds it on a machine tool to remove what is not needed; what is left at the end is the desired shape. This approach is called subtractive manufacturing.

3 3D打印正好相反，它用的是“加法”。打印机[1]不是从大块材料开始，而是一层一层地把材料加上去。每一层都很薄，常常只有零点一到零点三毫米。成百上千[2]层叠[3]在一起，就成了一个完整的物体[4]。所以，3D打印的正式名字叫做增材制造[5]。

4 这种“一层一层”的方法有一个很大的好处：形状再复杂，打印起来也不一定更难。用传统方法很难做出来的内部空心结构、弯弯曲曲[6]的管道[7]，3D打印都可以比较容易地完成。另外，因为只在需要的地方加材料，浪费也比较少。

1 打印机 *dǎyìnjī* printer
2 成百上千 *chéngbǎi shàngqiān* hundreds or thousands
3 叠 *dié* to stack
4 物体 *wùtǐ* object
5 增材制造 *zēngcái zhìzào* additive manufacturing
6 弯弯曲曲 *wānwānqūqū* winding, twisting
7 管道 *guǎndào* pipe, channel

3D printing is exactly the opposite: it works by "addition". Instead of start- 3
ing from a large block of material, the printer adds material one layer at a time. Each layer is very thin, often only 0.1 to 0.3 millimetres. Hundreds or thousands of layers stacked together make a complete object. That is why the formal name for 3D printing is additive manufacturing.

This "layer by layer" method has one big advantage: however complex the 4
shape, it is not necessarily any harder to print. Hollow internal structures and winding channels that are very hard to make by traditional methods can be produced fairly easily by 3D printing. And because material is added only where it is needed, there is also less waste.

从模型到打印机[1]

3D打印的第一步是设计模型。工程师一般用CAD软件[2]在电脑上画出物体[3]的三维模型[4]。如果不会设计，也可以从网上下载别人分享[5]的模型，或者用三维扫描仪[6]扫描[7]一个真实的东西。

设计好以后，模型常常被保存成STL格式[8]。STL文件把物体的表面分成很多很小的三角形，就像用很多小纸片拼[9]出一个形状。三角形越多，表面就越光滑[10]，但是文件也越大。

第二步叫做“切片[11]”。切片软件会把模型切成很多水平的薄层，然后计算出打印机在每一层应该怎么走、走多快、在哪里放材料。

1 打印机 *dǎyìnjī* printer
2 软件 *ruǎnjiàn* software
3 物体 *wùtǐ* object
4 三维 *sānwéi* three-dimensional
5 分享 *fēnxiǎng* to share
6 扫描仪 *sǎomiáoyí* 3D scanner
7 扫描 *sǎomiáo* to scan
8 格式 *géshì* file format
9 拼 *pīn* to piece together
10 光滑 *guānghuá* smooth
11 切片 *qiēpiàn* slicing

From Model to Printer

The first step in 3D printing is designing a model. Engineers usually draw a three-dimensional model of the object on a computer with CAD software. If you cannot design, you can also download models that others have shared online, or scan a real object with a 3D scanner. 5

Once the design is finished, the model is usually saved in STL format. An STL file divides the surface of the object into many tiny triangles, like piecing a shape together out of lots of little scraps of paper. The more triangles, the smoother the surface, but the bigger the file. 6

The second step is called “slicing”. The slicing software cuts the model into many thin horizontal layers and then works out how the printer should move in each layer, how fast, and where to put material. 7

7 用户[1]可以在软件[2]里设置很多参数[3]，比如层高[4]、打印速度、温度和填充率[5]。

8 最后，切片[6]软件会生成[7]G-code。G-code是一种很简单的机器语言，里面是一行一行的命令[8]，比如“移动到这个位置”、“把喷嘴[9]加热[10]到两百度”。打印机[11]按照这些命令一步一步地工作，几个小时以后，模型就从电脑里“走”到了桌子上。

常见的打印方法

9 最常见的一种方法叫FDM，中文叫熔融沉积成型[12]。家里和学校用的3D打印机大多是这一种。

1 用户 *yònghù* user
2 软件 *ruǎnjiàn* software
3 参数 *cānshù* parameter
4 层高 *cénggāo* layer height
5 填充率 *tiánchōnglǜ* infill percentage
6 切片 *qiēpiàn* slicing
7 生成 *shēngchéng* to generate
8 命令 *mìnglìng* command
9 喷嘴 *pēnzuǐ* nozzle
10 加热 *jiārè* to heat
11 打印机 *dǎyìnjī* printer
12 熔融沉积成型 *róngróng chénjī chéngxíng* fused deposition modelling

Users can set many parameters in the software, such as layer height, print speed, temperature and infill percentage. 7

Finally, the slicer generates G-code. G-code is a very simple machine language made up of one command per line, such as "move to this position" or "heat the nozzle to 200 degrees". The printer follows these commands step by step, and a few hours later the model has "walked" out of the computer and onto the table. 8

Common Printing Methods

The most common method is called FDM, which in Chinese is called "fused deposition modelling". Most 3D printers found in homes and schools are of this kind. 9

它的原理[1]有点像一把热熔胶枪[2]：一根塑料丝[3]被送进加热[4]的喷嘴[5]，在两百度左右熔化[6]，然后从很细的喷嘴口挤[7]出来。喷嘴在平台[8]上方[9]移动，一条线一条线地画出每一层，塑料很快就冷却变硬[10]了。

一台FDM打印机[11]的结构并不复杂。它主要有三个部分：负责送丝[12]和加热的挤出机[13]、可以前后、左右、上下[14]移动的运动系统，还有一块平台。运动系统一般由几个步进电机[15]控制[16]，电机[17]每走一步只转一个很小的角度，所以喷嘴的位置可以控制得很准确。打印完一层以后，平台下降一点点，或者喷嘴升高[18]一点点，然后开始打印下一层。

FDM打印机常用的材料有PLA、ABS和PETG。

1 原理 *yuánlǐ* principle
2 热熔胶枪 *rèróngjiāoqiāng* hot glue gun
3 塑料丝 *sùliàosī* plastic filament
4 加热 *jiārè* to heat
5 喷嘴 *pēnzuǐ* nozzle
6 熔化 *rónghuà* to melt
7 挤 *jǐ* to squeeze out
8 平台 *píngtái* platform, build plate
9 上方 *shàngfāng* above
10 硬 *yìng* hard
11 打印机 *dǎyìnjī* printer
12 丝 *sī* filament
13 挤出机 *jǐchūjī* extruder
14 上下 *shàngxià* up and down
15 步进电机 *bùjìn diànjī* stepper motor
16 控制 *kòngzhì* to control
17 电机 *diànjī* electric motor
18 升高 *shēnggāo* to rise

It works a little like a hot glue gun: a plastic filament is fed into a heated nozzle, melts at around 200 degrees and is squeezed out through the nozzle's fine opening. The nozzle moves above the platform, drawing each layer line by line, and the plastic quickly cools and hardens. 9

The construction of an FDM printer is not complicated. It has three main parts: the extruder, which feeds and heats the filament; the motion system, which moves forwards and backwards, left and right, up and down; and a platform. The motion system is usually driven by several stepper motors; with each step a motor turns through only a tiny angle, so the position of the nozzle can be controlled very accurately. When a layer is finished, the platform drops a tiny amount, or the nozzle rises a tiny amount, and the next layer begins. 10

Common materials for FDM printers are PLA, ABS and PETG. 11

11 PLA最容易打印，气味[1]也比较小，适合初学者；ABS更结实、更耐热[2]，但是打印的时候容易翘边[3]，对温度的要求也更高；PETG的性能[4]在两者之间。打印平台[5]一般也是加热[6]的，叫做热床[7]，这样第一层可以牢牢[8]地粘[9]在上面，不容易翘[10]起来。

12 打印的时候，物体[11]里面一般不是实心的，而是用一种网格一样的结构来填充[12]。填充率[13]越高，零件[14]就越结实，但是用的材料和时间也越多。还有一个问题是“悬空[15]”：如果某一部分下面什么都没有，熔化[16]的塑料就会掉下去。这时候，切片[17]软件[18]会自动加上支撑[19]，打印完以后再把支撑去掉[20]。一般来说，倾斜[21]不超过四十五度的部分可以不用支撑。

1 气味 *qìwèi* smell
2 耐热 *nàirè* heat-resistant
3 翘边 *qiàobiān* to warp at the edges
4 性能 *xìngnéng* properties, performance
5 平台 *píngtái* platform, build plate
6 加热 *jiārè* to heat
7 热床 *rèchuáng* heated bed
8 牢牢 *láoláo* firmly
9 粘 *zhān* to stick
10 翘 *qiào* to curl up
11 物体 *wùtǐ* object
12 填充 *tiánchōng* to fill; infill
13 填充率 *tiánchōnglǜ* infill percentage
14 零件 *língjiàn* part, component
15 悬空 *xuánkōng* overhang
16 熔化 *rónghuà* to melt
17 切片 *qiēpiàn* slicing
18 软件 *ruǎnjiàn* software
19 支撑 *zhīchēng* support (structure)
20 去掉 *qùdiào* to remove
21 倾斜 *qīngxié* to slant, to lean

PLA is the easiest to print and has little smell, so it suits beginners; ABS is 11
tougher and more heat-resistant, but tends to warp at the edges while printing and is fussier about temperature; PETG lies between the two. The build platform is usually heated as well — it is called a heated bed — so that the first layer sticks firmly to it and does not easily curl up.

When printing, the inside of an object is usually not solid but filled with a 12
grid-like structure. The higher the infill percentage, the stronger the part, but the more material and time it takes. Another issue is the "overhang": if there is nothing underneath some part, the molten plastic will simply drop down. In that case the slicer automatically adds supports, which are removed after printing. As a rule, parts that lean out by no more than 45 degrees can be printed without supports.

13 第二种常见的方法是光固化[1]，比如 SLA 和 DLP。这种打印机[2]用的不是塑料丝[3]，而是一种液体[4]树脂[5]，它遇到紫外线[6]就会变硬[7]。打印机用激光[8]或者投影一层一层地照射[9]树脂，被照到的地方就固化[10]成固体。光固化打印的精度[11]很高，表面也很光滑[12]，所以常常用来[13]做首饰[14]模型、牙齿[15]模型和很小的零件[16]。不过，没有固化的树脂对皮肤[17]有刺激，使用的时候要戴手套，还要注意通风[18]。

14 第三种是粉末[19]打印，比如 SLS，中文叫选择性激光烧结[20]。机器先在平台[21]上铺[22]一层很薄的粉末，然后用激光把需要的部分烧结[23]在一起，再铺下一层粉末，就这样一直重复。

1 光固化 *guānggùhuà* light curing (resin printing)
2 打印机 *dǎyìnjī* printer
3 塑料丝 *sùliàosī* plastic filament
4 液体 *yètǐ* liquid
5 树脂 *shùzhī* resin
6 紫外线 *zǐwàixiàn* ultraviolet light
7 硬 *yìng* hard
8 激光 *jīguāng* laser
9 照射 *zhàoshè* to shine on
10 固化 *gùhuà* to cure, to solidify
11 精度 *jīngdù* precision
12 光滑 *guānghuá* smooth
13 用来 *yònglái* to be used for
14 首饰 *shǒushì* jewellery
15 牙齿 *yáchǐ* teeth
16 零件 *língjiàn* part, component
17 皮肤 *pífū* skin
18 通风 *tōngfēng* ventilation
19 粉末 *fěnmò* powder
20 选择性激光烧结 *xuǎnzéxìng jīguāng shāojié* selective laser sintering
21 平台 *píngtái* platform, build plate
22 铺 *pū* to spread
23 烧结 *shāojié* to sinter

The second common method is light curing, as in SLA and DLP printers. 13
These printers use not filament but a liquid resin that hardens when it is exposed to ultraviolet light. The printer shines a laser or a projected image onto the resin layer by layer, and wherever the light falls the resin cures into a solid. Resin printing is very precise and the surface is smooth, so it is often used for jewellery models, dental models and very small parts. However, uncured resin irritates the skin, so you should wear gloves when handling it and make sure the room is well ventilated.

The third is powder printing, such as SLS — in Chinese, "selective laser sin- 14
tering". The machine first spreads a very thin layer of powder on the platform, then uses a laser to fuse the required areas together, spreads the next layer of powder, and keeps repeating.

14 没有被烧结[1]的粉末[2]可以托住[3]上面的部分，所以一般不需要另外加支撑[4]。SLS 常用尼龙[5]粉末。用类似的方法和更强的激光[6]，还可以把钛合金[7]、不锈钢[8]这样的金属[9]粉末熔化[10]成零件[11]，这在航空和医疗领域[12]特别有用。

优点和不足[13]

15 3D打印最大的优点是灵活[14]。做一个新零件，不需要先做模具[15]，只要改一下电脑里的模型，第二天就可以拿到新的样品[16]。所以，很多公司用它来做原型[17]，快速检查设计有没有问题。对于只需要做一两个的特殊零件，3D打印也常常比传统方法便宜得多。

1 烧结 *shāojié* to sinter
2 粉末 *fěnmò* powder
3 托住 *tuōzhù* to hold up
4 支撑 *zhīchēng* support (structure)
5 尼龙 *nílóng* nylon
6 激光 *jīguāng* laser
7 钛合金 *tàihéjīn* titanium alloy
8 不锈钢 *bùxiùgāng* stainless steel
9 金属 *jīnshǔ* metal
10 熔化 *rónghuà* to melt
11 零件 *língjiàn* part, component
12 领域 *lǐngyù* field, sector
13 不足 *bùzú* shortcoming
14 灵活 *línghuó* flexible
15 模具 *mújù* mould
16 样品 *yàngpǐn* sample
17 原型 *yuánxíng* prototype

The unsintered powder holds up the parts above it, so separate supports are generally not needed. SLS commonly uses nylon powder. With a similar method and a more powerful laser, metal powders such as titanium alloy or stainless steel can also be melted into parts, which is especially useful in aviation and medicine. 14

Strengths and Weaknesses

The greatest strength of 3D printing is its flexibility. To make a new part you do not have to make a mould first; you simply change the model on the computer and can have a new sample the next day. That is why many companies use it to make prototypes and quickly check whether a design has any problems. For special parts that are needed only once or twice, 3D printing is also often much cheaper than traditional methods. 15

16 但是，3D打印也有不少问题。首先是速度：打印一个手掌[1]大小的零件[2]，常常需要好几个小时，所以它不适合大量生产。其次，因为物体[3]是一层一层叠[4]起来的，层和层之间的连接[5]比较弱。如果力的方向和层平行，零件比较结实；如果力要把层和层拉开，零件就容易断。工程师把这种特点叫做各向异性[6]，所以设计的时候要考虑零件的打印方向。

17 另外，FDM打印的表面常常能看到一条一条的层纹[7]，不够光滑[8]，有时候需要打磨[9]或者再加工。打印失败也是常有的事：第一层没粘好[10]、喷嘴[11]堵住了、模型打到一半倒了……这些都会让几个小时的工作白费。

1 手掌 *shǒuzhǎng* palm (of the hand)
2 零件 *língjiàn* part, component
3 物体 *wùtǐ* object
4 叠 *dié* to stack
5 连接 *liánjiē* connection, bond
6 各向异性 *gèxiàngyìxìng* anisotropy
7 层纹 *céngwén* layer lines
8 光滑 *guānghuá* smooth
9 打磨 *dǎmó* to sand, to polish
10 粘 *zhān* to stick
11 喷嘴 *pēnzuǐ* nozzle

But 3D printing also has quite a few problems. The first is speed: printing 16
a palm-sized part often takes several hours, so it is not suited to mass production. Second, because the object is built up layer by layer, the bond between the layers is relatively weak. If a force acts parallel to the layers, the part is fairly strong; if the force tries to pull the layers apart, the part breaks easily. Engineers call this property anisotropy, so the print orientation of a part has to be considered at the design stage.

In addition, layer lines are often visible on the surface of FDM prints, so it is 17
not very smooth and sometimes needs sanding or further processing. Failed prints are common too: the first layer does not stick, the nozzle clogs, the model topples over halfway through... any of these can waste hours of work.

3D打印的应用

18 今天，3D打印已经走进了很多行业。在工厂里，工程师用它打印夹具[1]和工具；在医院里，医生可以根据病人的CT数据打印出骨头的模型，在手术前进行练习。用钛合金[2]打印的人工关节[3]和植入物[4]，还可以根据每个病人的身体专门设计。

19 在航空领域[5]，有的发动机零件[6]以前需要把很多小零件焊[7]在一起，现在可以一次打印成一个整体，不但更轻，而且更可靠。在建筑[8]领域，有的公司用巨大的打印机[9]把混凝土[10]一层一层地挤[11]出来，几天之内[12]就能“打印”出一座小房子的墙。

20 3D打印也受到了普通人的欢迎。

1 夹具 *jiājù* jig, fixture
2 钛合金 *tàihéjīn* titanium alloy
3 关节 *guānjié* joint
4 植入物 *zhírùwù* implant
5 领域 *lǐngyù* field, sector
6 零件 *língjiàn* part, component
7 焊 *hàn* to weld
8 建筑 *jiànzhù* construction
9 打印机 *dǎyìnjī* printer
10 混凝土 *hùnníngtǔ* concrete
11 挤 *jǐ* to squeeze out
12 之内 *zhīnèi* within

What 3D Printing Is Used For

Today 3D printing has found its way into many industries. In factories engineers use it to print jigs, fixtures and tools; in hospitals doctors can print models of bones from a patient's CT data and practise before an operation. Artificial joints and implants printed in titanium alloy can even be designed specifically for each patient's body. 18

In aviation, some engine parts that used to be welded together from many small pieces can now be printed in one go as a single piece, which is both lighter and more reliable. In construction, some companies use huge printers to extrude concrete layer by layer, and can "print" the walls of a small house within a few days. 19

3D printing has also become popular with ordinary people. 20

很多人在家里打印手机支架、玩具和坏了的家电[†]零件[‡]，还有人在网上免费分享[§]自己的设计。现在，一台入门级的打印机[¶]只要一两千元人民币就能买到。

在学校里，3D打印机也是一种很好的教学工具。学生可以自己设计一个小东西，第二天就拿到手里，看看哪里设计得好，哪里还需要修改。这种“设计、打印、检查、再设计”的过程，正是工程师每天工作的方式。

结语

3D打印不会马上代替所有传统的制造方法，因为在大量生产的时候，注塑[*]和机床[#]加工仍然更快、更便宜。但是，它让设计和制造之间的距离变得更短了。

† 家电 *jiādiàn* household appliance
‡ 零件 *língjiàn* part, component
§ 分享 *fēnxiǎng* to share
¶ 打印机 *dǎyìnjī* printer
* 注塑 *zhùsù* injection moulding
机床 *jīchuáng* machine tool

Many people print phone stands, toys and spare parts for broken household appliances at home, and some share their designs online for free. Nowadays an entry-level printer can be bought for just one or two thousand yuan. 20

In schools, too, the 3D printer is an excellent teaching tool. Students can design a small object themselves, hold it in their hands the next day, and see which parts are well designed and which still need changing. This cycle of "design, print, check, redesign" is exactly how engineers work every day. 21

In Closing

3D printing will not replace all traditional manufacturing methods any time soon, because for mass production, injection moulding and machining are still faster and cheaper. But it has shortened the distance between design and manufacturing. 22

22 一个想法从电脑里的模型变成拿在手里的东西，可能只需要一个晚上。也许将来，“打印”一件东西会像今天打印一张纸一样普通。

An idea can go from a model on a computer to something you hold in your hand in a single night. Perhaps one day "printing" an object will be as ordinary as printing a sheet of paper is today. 22